

An ISO 9001:2008 Certified

Crisp System

Arduino Kit FPGA Trainer Kit

DSP Trainer Fingerprint Module

Force Sensor Color Sensor

GSM Modem Voice Recognition Kit

MP3 Module Versa Pad

RF Transceiver Wi-Fi Transceiver

Accelerometer Sensor Module Thermoelectric Power Generator

Fire Sensor Potter Device

Humidity Sensor Module Pulse Rate Sensor Module

Voice Recording and Playback Module Power line Communication Module

RFID Reader Module Gas Sensor Module

All the above Modules are Ready to Interface with any Micro controller

Crisp Towers, 120, Nehru Street, Ramnagar
Coimbatore - 641 009
Cell : 7373 63 0000 / 98422 18988
Email : sales@crispprojects.com
Web : www.crispprojects.com

Project Guidance also available

trode. Interference signals include the 50/60Hz interference from the power supplies, motion artifacts due to patient movement, radio frequency interference from electro-surgery equipment, defibrillation pulses, pacemaker pulses, signals from other monitoring equipment, etc.

A smart designer would rather focus on the simplicity of the analogue signal acquisition than on display/print of the ECG graph as that determines the accuracy of the test equipment. Some of the 50Hz/60Hz common-mode interference can be cancelled with a high-input-impedance instrumentation amplifier, which removes the AC line noise common to both the inputs. To further reject line power noise, the signal is inverted and driven back into the patient through the right leg by an amplifier. Only a few microamps or less are required to achieve significant common-mode rejection (CMR) improvement and stay within the UL544 (UL Standard for Safety Medical and Dental Equipment) limit. In addition, 50/60Hz digital notch filters are used to reduce this interference further.

High levels of integration and cost reductions are achieved by using the new analogue front-end parts for the application. These help designers reduce the overall size of the board as well as power consumption, thereby making the machine portable and low-cost. This surely makes the life of doctors and patients simpler and safer while providing far greater access to technology and medical diagnostics.

Another example of a consumer product that is finding wide acceptance amongst consumers of varied ages, needs and interests is the tablet computer. Tablet is a hand-held electronic device primarily designed to run multimedia and Web-based applications that require considerable processing power. Its main advantage over a laptop PC or a notebook PC is its portability, which is achieved by having a small form factor and less number of peripheral devices. Additionally, it uses a touchscreen as the primary

interface to control and interact with the product.

Again, apart from the heart of this product, i.e., the applications processor, designers have realised the importance of focusing on analogue for differentiating their products from competition and offering a rich user experience. This is evident from the use of light sensors in displays to continuously monitor the ambient light and alter the brightness of the display in order to conserve the battery power, thereby enabling longer hours of use.

Semiconductor players provide designers with solutions for the LCD/OLED displays, which include low-power, high-ESD protection touch-screen controllers, display interface products, and LCD bias and backlight solutions. They provide complete power management solution for tablet design, including a power management unit (PMU) integrated with a USB-OTG transceiver and a keypad controller. The outstanding efficiency of power conversion extends battery run-time in addition to minimising heat dissipation. Accurate fuel gauging maximises usable battery capacity and ensures orderly shutdown. The option to choose between system-side and pack-side battery fuel gauge offers designers flexibility in design. Similarly, designers using low-power audio codecs that integrate stereo headphone and class-D speaker amplifiers are able to extend battery life.

All the above focuses on solving the most critical problem faced by designers, i.e., battery life, via the analogue way!

To sum up

While the world moves towards higher throughputs and speed with faster processors, the focus of designers to provide simple, efficient and safe products will still have to remain on analogue design, as the only way to interact with humans is analogue! ●

The author is regional sales manager-Gurgaon and Chandigarh, Texas Instruments (India)